Supplementary Information

Behavioral innovation enables the poleward range expansion of a tropical ectotherm beyond its physiological limits

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All code and data used to produce these materials are available at https://github.com/ggcostoya/green-anole-litter, and every analysis can be explored at https://ggcostoya.github.io/green-anole-litter/.

Supplementary Note 1. Biophysical models

Here we describe in detail how we estimated the minimum annual body (i.e., operative) temperatures that an American green anole (*Anolis carolinensis*) would experience in the canopy and at different depths in the leaf litter across southeastern North America, which we used in the mechanistic envelope models (Supplementary Fig. 9). All models were implemented as vectorized functions in R, and the code is available at the project's GitHub repository and website.

1. Range of the green anole

We downloaded all research-grade iNaturalist observations of the green anole and curated them by (i) rounding coordinates to one decimal place, (ii) removing observations with a positional uncertainty greater than 1 km, (iii) keeping observations within the native range of the species (longitudes between 110° and 70°W and latitudes between 22.5° and 37.5°N), (iv) counting observations within each 0.1° × 0.1° cell and keeping only cells with at least three observations, to eliminate isolated observations caused by potential errors or rare occurrences, and (v) removing cells with fewer than three other occupied cells within 50 km. We then manually drew a polygon around the curated cells to define the range of the species (Supplementary Fig. 10), and assumed that the green anole is present in all cells within this polygon and absent from all cells outside of it.

2. Environmental data

For every 2.5′ latitude-longitude cell within a study area extending 2° beyond the range polygon in all directions, and restricted to land areas of the continental United States and Mexico, we obtained:

- **Forest cover** (C, from 0 to 1): the proportion of each cell covered by trees, mangroves or wetlands in the ESA WorldCover dataset (accessed through the `landcover` function of the R package `geodata`), aggregated from its original 30″ resolution to 2.5′.
- **Elevation** ($Elev$, in m) from WorldClim 2.1[1].
- **Minimum and maximum air temperatures** (T_{min} and T_{max}, in °C) of January, the coldest month of the year, from WorldClim 2.1.

3. Microclimate models

For each cell, we generated hourly profiles of solar radiation and temperature on a representative day of the coldest month of the year (15 January, day of the year $DOY = 15$), and used them to estimate temperatures at 23 heights and depths: 1.8 and 2 m above the ground (the canopy), the ground surface, and leaf-litter depths from 1 to 20 cm at 1-cm intervals. All model parameters are listed in Supplementary Table 11.

3.1. Solar radiation

We estimated hourly direct solar radiation (R_{solar}) as a function of location, day of the year and elevation. We first estimated the ecliptic longitude of the sun (λ) as a function of the day of the year and the eccentricity of the Earth's orbit (e):

$$\lambda = \frac{2\pi}{365}(DOY - 80) + 2e\left[\sin\left(\frac{2\pi}{365}DOY\right) - \sin\left(\frac{2\pi}{365}80\right)\right]$$

where day 80 approximates the vernal equinox. From this, we calculated the solar declination angle (δ):

$$\delta = \arcsin(0.3879672 \cdot \sin\lambda)$$

where $\arcsin(0.3879672) = 22.82°$ approximates the Earth's average axial tilt over a year. The sine of the solar altitude angle at hour h ($a(h)$) was then calculated from latitude (lat) and the time of solar noon (S_n, obtained with the `solar_noon` function of the R package `TrenchR`[2]):

$$a(h) = \sin(lat)\sin(\delta) + \cos(lat)\cos(\delta)\cos\left[\,0.2618\,(h - S_n)\,\right]$$

where 0.2618 rad h^{-1} is the Earth's rotational speed (15° h^{-1}). To account for the effect of elevation on the atmospheric absorption of solar radiation, we modeled barometric pressure (p_a, in kPa) as:

$$p_a = 101.3\, e^{-Elev/8200}$$

and calculated the atmospheric path length (m_a) as:

$$m_a(h) = \frac{p_a}{101.3\, a(h)}$$

Finally, hourly direct solar radiation (in W m^{-2}) was calculated from the extraterrestrial solar constant (E_{sc}) and the atmospheric transmittance coefficient (τ), and set to zero when the sun is below the horizon:

$$R_{solar}(h) = 1000 \cdot E_{sc}\, \tau^{m_a(h)}\, a(h)$$

3.2. Air temperature

We estimated hourly air temperature (T_{air}) by interpolating between the daily minimum and maximum air temperatures with a sinusoidal weighting function[3]. With the angular frequency of the daily temperature cycle $\omega = \pi/12$ rad h^{-1}, the weight of the maximum temperature at hour h is:

$$\gamma(h) = 0.44 - 0.46\sin(0.9 + \omega h) + 0.11\sin(0.9 + 2\omega h)$$

and hourly air temperature is:

$$T_{air}(h) = T_{max}\, \gamma(h) + T_{min}\, [1 - \gamma(h)]$$

3.3. Effect of forest cover on air temperature

Forest canopies intercept part of the incoming solar radiation, cooling the air below them during the day. We modeled this effect as proportional to both solar radiation and forest cover, using a canopy cooling coefficient (b_c):

$$T_{air}^{'}(h) = T_{air}(h) - b_c\, R_{solar}(h)\, C$$

Under peak solar radiation (800–1000 W m^{-2}) and a fully closed canopy, this coefficient yields a maximum daytime cooling of 5.6–7 °C, consistent with measurements from understory microclimate logger networks in temperate and subtropical forests[4–6].

3.4. Surface temperature

Exposed ground surfaces heat up above air temperature during the day. We modeled this effect as proportional to solar radiation and to the proportion of the ground not covered by forest, using a surface heating coefficient (b_s):

$$T_{surf}(h) = T_{air}^{'}(h) + b_s \, R_{solar}(h) \, (1 - C)$$

Under peak unshaded solar radiation, this coefficient yields a daytime surface temperature 16–20 °C above air temperature, consistent with standard microclimate energy balance formulations[3] and measurements of forest floor microclimates[5,7].

3.5. Temperatures above the surface

Above the surface, we assumed a logarithmic transition between surface temperature and the forest-corrected air temperature at the reference height of the climate data ($z_{ref} = 1.8$ m). The relative influence of air temperature at height z was described by:

$$\phi(z) = \frac{\ln\left[\,(z + z_0)/z_0\,\right]}{\ln\left[\,(z_{ref} + z_0)/z_0\,\right]}$$

where z_0 is a small roughness length that avoids the singularity of the logarithm at the surface. Temperature at height z was then:

$$T(z, h) = T_{surf}(h) - \left[T_{surf}(h) - T_{air}^{'}(h)\right] \phi(z)$$

3.6. Temperatures in the leaf litter

Below the surface, we modeled the propagation of the daily temperature cycle into the leaf litter using the analytical solution of the one-dimensional heat conduction equation[3]. The thermal diffusivity of the leaf litter (κ) was calculated from its thermal conductivity (k), bulk density (ρ) and specific heat capacity (c_p):

$$\kappa = \frac{k}{\rho \, c_p}$$

We used values representative of dry, loosely packed organic leaf litter rather than mineral soil, since the high porosity of litter (80–90%) traps insulating pockets of still air that give it a much lower thermal conductivity and bulk density than mineral soil (Supplementary Table 11). From the diffusivity and the angular frequency of the daily cycle ($\omega = 2\pi/24$ rad h^{-1}), we calculated the damping depth (D), at which the amplitude of the surface temperature cycle is reduced by a factor of e:

$$D = \sqrt{2\kappa/\omega}$$

At depth z, temperature fluctuations are reduced in amplitude by a factor $A(z)$ and delayed by $\Delta t(z)$ hours relative to the surface:

$$A(z) = e^{-|z|/D} \qquad \Delta t(z) = \frac{|z|}{D} \cdot \frac{24}{2\pi}$$

Temperature at depth z was then calculated by damping the delayed surface temperature (obtained by linear interpolation of the hourly surface temperature profile, treating it as a repeating daily cycle)

around the mean daily surface temperature ($\bar{T}_{surf}$):

$$T(z,h) = \bar{T}_{surf} + \left[T_{surf}(h - \Delta t(z)) - \bar{T}_{surf}\right] A(z)$$

For each cell and microclimate, we retained the minimum hourly temperature of the day as the minimum annual temperature of that microclimate.

4. Operative temperatures

We transformed minimum annual microclimate temperatures into the minimum operative temperatures (T_e) of a green anole, i.e., the steady-state body temperature a lizard reaches in thermal equilibrium with its environment. Because the coldest temperatures of the day occur at night, we assumed that there was no solar radiation.

Within the leaf litter, a lizard is fully surrounded by litter at ambient temperature and has no view of the night sky, so we assumed that its operative temperature equals the temperature of the litter at that depth, $T_e = T(z)$.

Above the ground, we calculated operative temperature as the conductance-weighted average of the minimum air temperature in that microclimate ($T(z)$), ground temperature (T_{ground}, assumed equal to air temperature) and an equivalent radiant temperature (T_{rad}):

$$T_e = \frac{h_c\, T(z) + h_k\, T_{ground} + h_r\, T_{rad}}{h_c + h_k + h_r}$$

where h_c, h_k and h_r are the convective, conductive and radiative heat transfer coefficients (W m^{-2} K^{-1}). We scaled body mass (m, in kg; 3.1 g, the typical mass of a green anole) to a characteristic length (L, in m) assuming standard reptile allometry, $L = 0.02\, m^{1/3}$, and calculated the convective heat transfer coefficient as the larger of forced convection (driven by wind speed, u) and free convection:

$$h_c = \max\left(\frac{10.3\, u^{0.6}}{L^{0.4}},\ 3.5\, L^{-0.25}\right)$$

We set wind speed to 0.1 m s^{-1}, representing the near-still air typical of sheltered forest understories at night, which minimizes convective heat exchange and thereby amplifies the relative effect of radiative cooling. Conduction to the perch was represented by a fixed coefficient (h_k) representing a small lizard resting against a solid substrate. Radiative heat transfer was linearized around air temperature (in K) using the Stefan-Boltzmann constant (σ) and the emissivity of lizard skin (ϵ_{lizard}):

$$h_r = 4\, \epsilon_{lizard}\, \sigma\, T_{air,K}^3$$

The equivalent radiant temperature combines thermal radiation from the night sky and from the ground, weighted by their view factors (F_{sky} and F_{ground}; 0.5 each for a lizard sleeping horizontally on a perch):

$$T_{sky,K} = \epsilon_{sky}^{0.25}\, T_{air,K} \qquad T_{rad,K} = \left(F_{sky}\, T_{sky,K}^4 + F_{ground}\, T_{ground,K}^4\right)^{0.25}$$

where the emissivity of a clear night sky (ϵ_{sky}) makes its effective temperature substantially lower than air temperature. These assumptions are simplifications of real-world conditions, but they provide a conservative (i.e., cold) estimate of the operative temperature of a small lizard sleeping in the canopy on a clear, calm winter night.

5. Validation

We validated the biophysical models by comparing the predicted minimum operative temperatures in the canopy (1.8 m) and in the leaf litter (8 cm, the midpoint of the 5–10 cm depths at which litter OTMs were deployed) of the cells within 0.1° of each of the five locations where we deployed operative temperature models (OTMs) with the mean daily minimum temperatures recorded by each OTM between 11 and 19 January, a window centered on the representative day used in the models. Predictions closely matched observations in both microhabitats (Supplementary Fig. 8 and Supplementary Table 8).

Supplementary Figures

Supplementary Fig. 1 | Green anoles retreat into leaf litter and other ground-level refuges during cold weather. Photographs and observer notes, reported by iNaturalist users, of American green anoles (*Anolis carolinensis*) that had retreated into the leaf litter or under other ground-level cover during cold episodes across the species' temperate range.



Oakman, Georgia, USA, late November 2020. Credit to Flown Kimmerling (https://www.inaturalist.org/observations/11361735): "In deep leaf litter in a mixed forest understory."



Lewisville, Texas, USA, during a cold snap in April 2025. A green anole (zoomed in on the left) was found next to a Texas spiny lizard (*Sceloporus olivaceus*; right). Credit to Daric Rodgers (https://www.inaturalist.org/observations/269032085): "A Green Anole and a Texas Spiny Lizard found side-by-side attempting to survive an April cold snap that got down into the 30s [~ -1 °C]. I removed a landscaping paver that provided a roof over some compacted leaf litter that was wedged against the wall right beside my exterior dryer vent."





Garland, Texas, USA, November 2019. Credit to Kala King (top: https://www.inaturalist.org/observations/100186678; bottom: https://www.inaturalist.org/observations/99989775): (top) “I found this cold anole by carefully scuffing through the leaf litter. It crawled out from under a pile of leaves. And kept slowly crawling until it disappeared under another pile of leaves.” (bottom) “Under a small log I found an anole asleep. In cold weather they stay under leaves and wood, don’t eat and don’t move very much.”



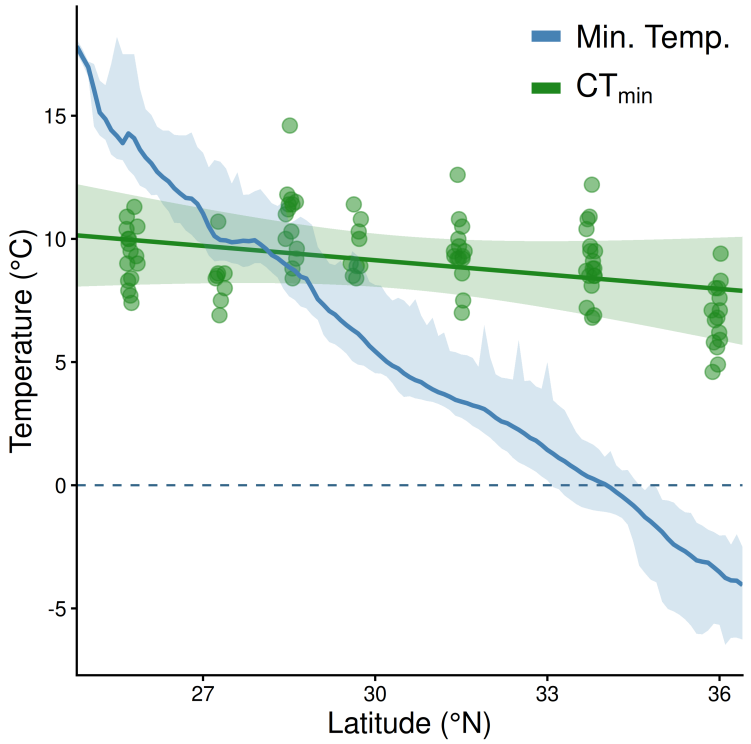
Oakman, Georgia, USA, April 2018. Credit to Flown Kimmerling (https://www.inaturalist.org/observations/11361735): “Under deep leaf litter on the side of a ridge near a seasonal stream in a dense mixed hardwood/coniferous forest in NW Georgia (Gordon County), US.”



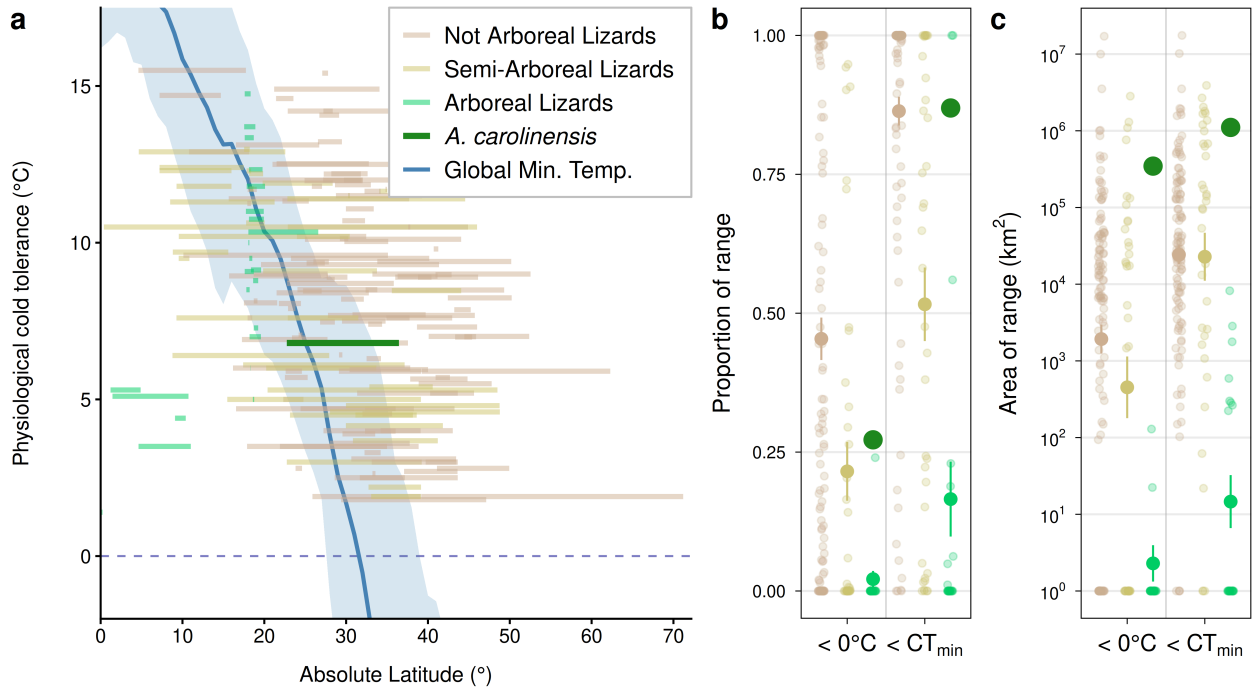
Spartanburg, South Carolina, USA, March 2021. Credit to Sharon Watson (https://www.inaturalist.org/observations/71590310): “Lethargic. Cold?”



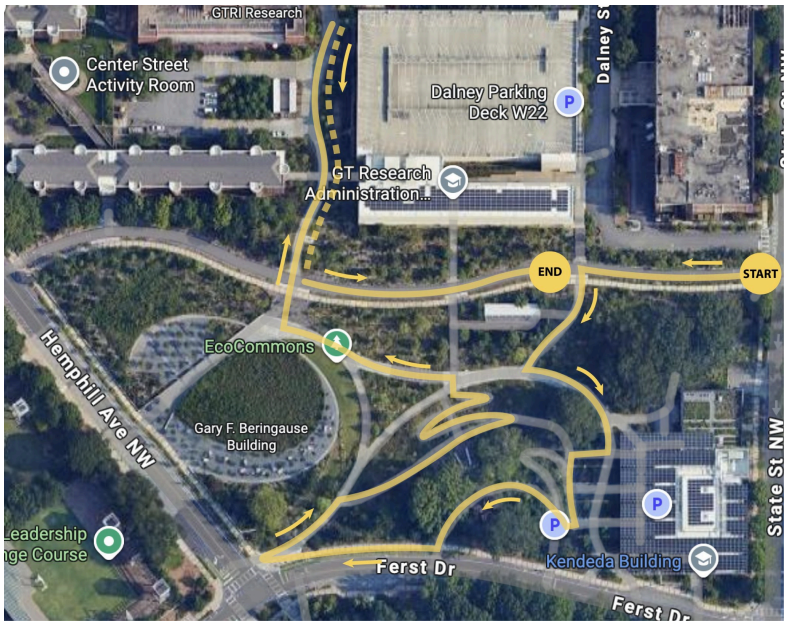
Columbia, South Carolina, USA, February 2025. Credit to Grant Foster (https://www.inaturalist.org/observations/261977666): “Found hanging out underneath a rotting log, hiding from the cold.”



Supplementary Fig. 2 | Latitudinal variation in the minimum temperature of the coldest month (BIO6, shown as Min. Temp.; blue) and the critical thermal minimum (CT_{min}; green) of American green anoles (*Anolis carolinensis*). For BIO6, the solid line and shaded area show the mean and range (minimum to maximum) of values within each 0.1° latitude band of a bounding box defined by the locations where we measured CT_{min} (extended by 1° to the east, west and south, and 0.5° to the north). For CT_{min}, points show individual estimates (n = 95 lizards from 7 populations; jittered horizontally for visibility), and the solid line and shaded area show the predicted latitudinal trend and its 95% confidence interval from a linear mixed model with population as a random effect (Supplementary Table 2). The dashed line marks 0 °C.



Supplementary Fig. 3 | The American green anole (*Anolis carolinensis*) is an exceptional case of mismatch between realized and fundamental cold thermal limits among arboreal lizards. a, The latitudinal range of each of the 180 lizard species in our dataset, placed at the height of its critical thermal minimum (CT_{min}) and colored by substrate use (exclusively arboreal, semi-arboreal or other), compared with the global mean (± SD) minimum temperature of the coldest month (BIO6; blue line and area) at each latitude. The range of the green anole (dark green) extends into latitudes where BIO6 is well below its CT_{min} (6.8 °C, the lowest population-average CT_{min} across its range). **b**,**c**, Proportion (**b**) and area (**c**, log scale) of each species' range below its CT_{min} and below 0 °C. Points with error bars show the mean (± SE) of each substrate group, and the large dark green point the green anole. Only a few other exclusively arboreal species have ranges that extend below their CT_{min} (four *Chamaeleo* and four *Anolis* species), and only two below 0 °C (*Chamaeleo hoehnelii* and *C. schubotzi*); in all cases, the area affected is orders of magnitude smaller than for the green anole. Substrate use follows SquamBase, except for *Liolaemus tenuis*, which we classified as semi-arboreal because it adopts a saxicolous/terrestrial strategy in many parts of its range.

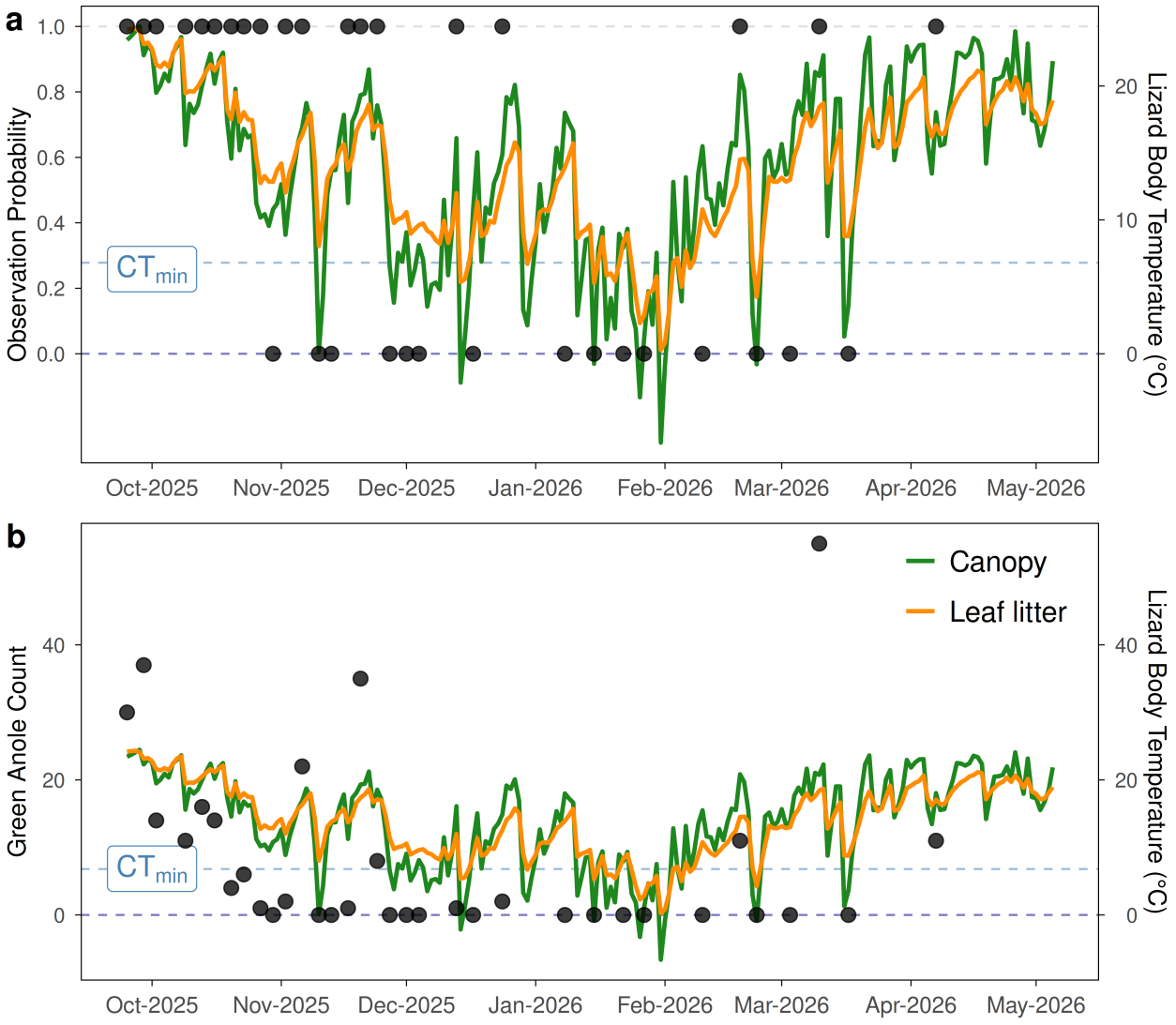




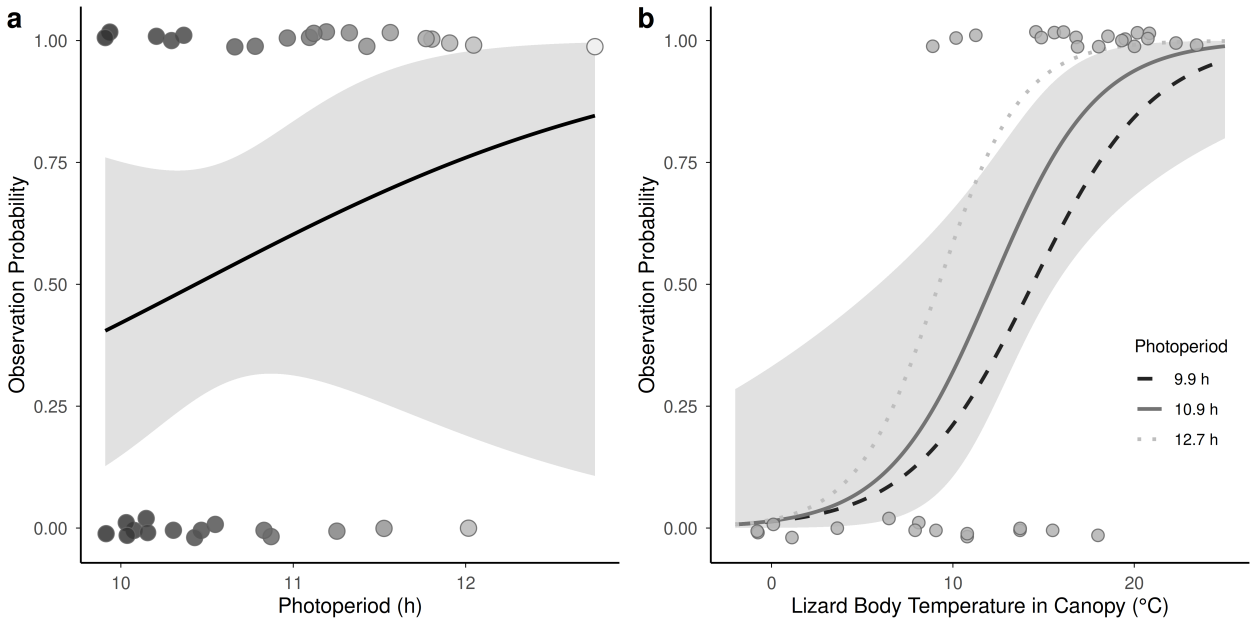
Supplementary Fig. 4 | Night survey transect on the Georgia Institute of Technology campus in Atlanta, Georgia, USA (top), and photographs of American green anoles (*Anolis carolinensis*) observed sleeping in vegetation during our surveys (bottom).



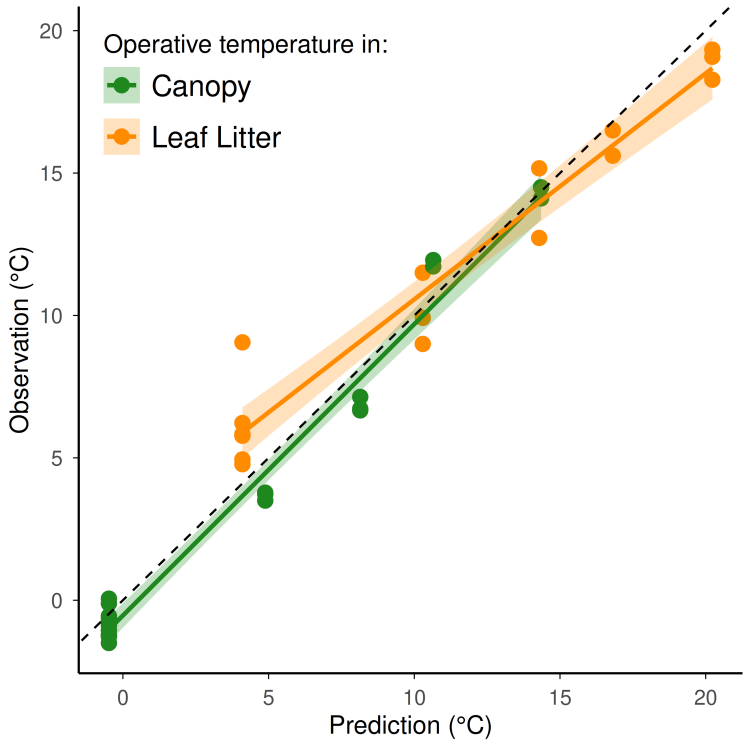
Supplementary Fig. 5 | Comparison between a perching American green anole (*Anolis carolinensis*; top) and an operative temperature model (OTM; bottom).



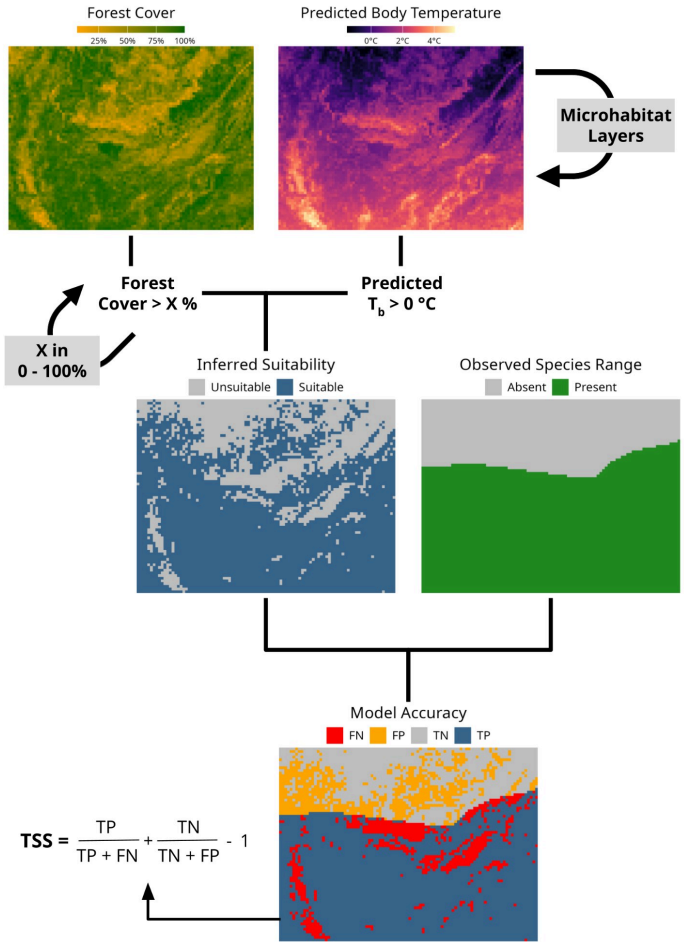
Supplementary Fig. 6 | Green anoles retreat into the leaf litter as nights cool. Presence (**a**) and number (**b**) of American green anoles (*Anolis carolinensis*) perching in the vegetation during our 34 night surveys between September 2025 and April 2026 in Atlanta, Georgia, USA (black points). Green and orange lines show the average operative temperature recorded by operative temperature models (OTMs) deployed in the canopy and in the leaf litter, respectively, between 1 and 2 hours after sunset on each day. The steel blue dashed line marks CT_{min} (6.8 °C, the lowest population-average CT_{min} across the species' range) and the dark blue dashed line 0 °C. As temperatures dropped in fall and winter, lizards abandoned the vegetation and retreated into the leaf litter, the only refuge available at the site, and they returned to the vegetation on warm nights in midwinter and spring, showing that retreat is a reversible response to temperature rather than a seasonal shift.



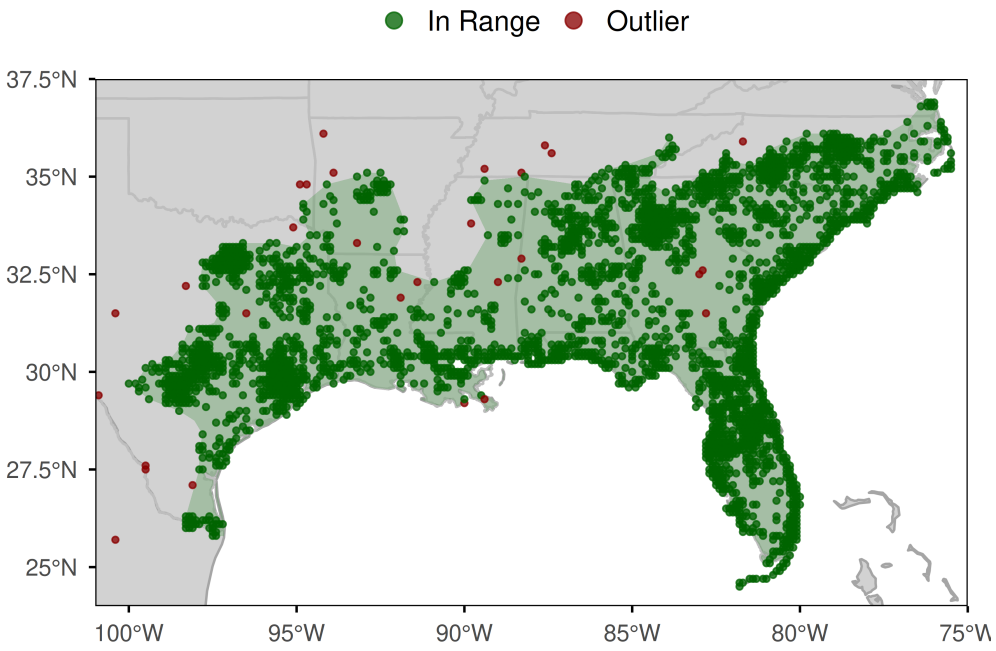
Supplementary Fig. 7 | Temperature, not photoperiod, triggers the retreat of American green anoles (*Anolis carolinensis*) into the leaf litter. Observation probability is the probability of observing lizards in the vegetation during a night survey, that is, before they retreat into the leaf litter. **a**, Predicted observation probability as a function of photoperiod at the average canopy operative temperature across surveys (13.1 °C). Points show the presence or absence of lizards in each survey, shaded by photoperiod (darker for shorter days). **b**, Predicted observation probability as a function of canopy operative temperature at the shortest (9.9 h), average (10.9 h) and longest (12.7 h) photoperiods of our surveys. Lines show the predictions of a Bernoulli generalized linear model including canopy temperature, photoperiod and their interaction (Supplementary Table 5), and shaded areas the 95% confidence intervals of the predictions (in **b**, at the average photoperiod only).



Supplementary Fig. 8 | Validation of the minimum operative temperatures estimated by our biophysical models against those recorded by operative temperature models (OTMs) at five locations across the latitudinal range of the American green anole (*Anolis carolinensis*). Each point shows the mean daily minimum temperature recorded by one OTM between 11 and 19 January (observation) against the minimum operative temperature predicted for its location and microhabitat (prediction; canopy at 1.8 m and leaf litter at 8 cm, the midpoint of the depths at which litter OTMs were deployed). Point colors indicate the microhabitat, and solid lines and shaded areas the best-fit line and standard error of a linear model for each microhabitat. The dashed black line indicates the 1:1 relationship. Predictions closely matched observations in the canopy (bias = 0.45 °C; RMSE = 0.88 °C; n = 22 OTMs) and in the leaf litter (bias = -0.41 °C; RMSE = 1.76 °C; n = 17 OTMs; Supplementary Table 8).



Supplementary Fig. 9 | Diagram of our mechanistic envelope modeling approach. We start by obtaining spatially explicit data on a set of ecological variables of interest, in our case forest cover and environmental temperature, and use both layers to derive new layers of predicted lizard body temperatures across a range of microhabitats using biophysical models (Supplementary Note 1). We then select a given microhabitat layer and set a percentage of forest cover as a threshold. Combining these two layers, we create an inferred suitability layer, assigning as suitable all spatial cells where forest cover is greater than the selected threshold and where the predicted minimum annual body temperature of the lizard (T_b) is greater than 0 °C. Next, we compare the inferred suitability layer with the observed species range cell by cell, classifying each cell as a false negative (FN; red) if we infer it unsuitable but it falls within the species range, a false positive (FP; orange) if we infer it suitable but it falls outside the range, a true negative (TN; gray) if we infer it unsuitable and it falls outside the range, or a true positive (TP; blue) if we infer it suitable and it falls within the range. Lastly, we calculate the true skill statistic (TSS) of each combination of microhabitat and forest cover threshold. We repeat this process for every microhabitat layer and every forest cover threshold (0 to 100% at 5% intervals) within a given area, and identify the combination that maximizes TSS.



Supplementary Fig. 10 | Range of the American green anole (*Anolis carolinensis*) used in our analyses. We used 0.1° latitude-longitude cells with at least three accurately georeferenced (positional uncertainty < 1 km) iNaturalist observations of the species (dark green points) within southeastern North America to define a polygon bounding the green anole's range (light green area). Cells with fewer than three other occupied cells within 50 km were considered outliers (red points) and were not considered when drawing the polygon.

Supplementary Tables

Supplementary Table 1 | Critical thermal minimum (CT_{min}) of American green anoles (*Anolis carolinensis*) and minimum temperature of the coldest month (BIO6) by location and latitude. CT_{min} values are the mean and standard deviation (SD) across individuals at each location. BIO6 values are the mean, minimum and maximum within the 0.1° latitude band containing each location, across a bounding box defined by all sampling locations (extended by 1° to the east, west and south, and 0.5° to the north).

Site	Latitude (°N)	N	Mean CT_{min} (°C)	SD CT_{min} (°C)	Mean BIO6 (°C)	Min BIO6 (°C)	Max BIO6 (°C)
Miami, FL	25.76	16	9.34	1.17	14.09	12.56	17.50
Port Saint Lucie, FL	27.30	8	8.40	1.11	9.96	9.12	13.13
Orlando, FL	28.54	14	10.77	1.57	8.85	7.67	10.63
Gainesville, FL	29.65	9	9.58	1.08	6.16	4.90	9.30
Tifton, GA	31.46	15	9.46	1.31	3.39	2.74	5.31
Atlanta, GA	33.75	18	9.05	1.42	0.35	-0.95	1.76
Knoxville, TN	35.95	15	6.80	1.33	-3.33	-5.62	-1.90

Supplementary Table 2 | Results of a linear mixed model examining how the CT_{min} of American green anoles (*Anolis carolinensis*) and the minimum temperature of the coldest month (BIO6) change with latitude. The intercept represents BIO6 at a latitude of zero, and C is an indicator variable for the temperature type (C = 0 for BIO6 and C = 1 for CT_{min}). The model includes a random intercept for population that applies only to CT_{min} observations, and a separate residual variance for each temperature type, and was fitted by restricted maximum likelihood (REML) with the R package `nlme`. Denominator degrees of freedom (df) for terms involving CT_{min} are based on the number of populations. The latitudinal slopes of each temperature type come from the same model, reparameterized with a separate intercept and slope for each type.

Parameter estimates

Parameter	Symbol	Estimate	Std. Error	df	t	p-value
Intercept	α	57.004	0.525	203	108.59	< 0.0001
Latitude	β_L	-1.690	0.017	203	-99.10	< 0.0001
C	β_C	-42.048	3.827	5	-10.99	0.0001
Latitude × C	$\beta_{L \cdot C}$	1.496	0.125	5	11.95	< 0.0001

Latitudinal slopes by temperature type

Temperature type	Slope (°C per ° latitude)	Std. Error	95% CI	df	t	p-value
BIO6 (β_L)	-1.690	0.017	-1.723 to -1.656	204	-99.10	< 0.0001
CT_{min} ($\beta_L + \beta_{L \cdot C}$)	-0.194	0.124	-0.513 to 0.125	5	-1.57	0.1782

Random effects and variance components

Component	Symbol	SD
Population (CT_{min} only)	σ_P	1.039
Residual, CT_{min}	$\sigma_{C=1}$	1.321
Residual, BIO6	$\sigma_{C=0}$	0.623

Model information

Observations	212 (95 CT_{min} from 7 populations; 117 BIO6 latitude bands)
Log-likelihood (REML)	-283.84
AIC (REML)	581.7

Supplementary Table 3 | Proportion (Prop.) and area of the range of 180 lizard species that falls below their CT_{min} and below 0 °C (based on the minimum temperature of the coldest month, BIO6), together with their CT_{min} and substrate use. All areas are expressed in millions of km^2. Proportions could not be calculated for two species whose ranges contain no climate grid cells within their elevational limits. The American green anole (*Anolis carolinensis*) is highlighted.

Family	Genus	Species	Proportion < CT_{min}	Proportion < 0 °C	Area < CT_{min} (10^6 km^2)	Area < 0 °C (10^6 km^2)	Total area (10^6 km^2)
Teiidae	*Ameiva*	*festiva*	0.05	0.00	0.0247	0.0000	0.4520
Agamidae	*Amphibolurus*	*muricatus*	0.86	0.16	0.0975	0.0186	0.1128
Dactyloidae	*Anolis*	*aliniger*	0.00	0.00	0.0000	0.0000	0.0119
Dactyloidae	*Anolis*	*alumina*	1.00	0.00	0.0006	0.0000	0.0006
Dactyloidae	*Anolis*	*armouri*	1.00	0.00	0.0001	0.0000	0.0001
Dactyloidae	*Anolis*	*bahorucoensis*	0.43	0.00	0.0001	0.0000	0.0001
Dactyloidae	*Anolis*	*baleatus*	0.00	0.00	0.0000	0.0000	0.0207
Dactyloidae	*Anolis*	*barahonae*	0.00	0.00	0.0000	0.0000	0.0003
Dactyloidae	*Anolis*	*barbouri*	0.24	0.00	0.0018	0.0000	0.0076
Dactyloidae	*Anolis*	*brevirostris*	0.00	0.00	0.0000	0.0000	0.0089
Dactyloidae	*Anolis*	*carolinensis*	0.87	0.27	1.1063	0.3463	1.2725
Dactyloidae	*Anolis*	*chlorocyanus*	0.00	0.00	0.0000	0.0000	0.0109
Dactyloidae	*Anolis*	*christophei*	0.19	0.00	0.0003	0.0000	0.0014
Dactyloidae	*Anolis*	*coelestinus*	0.23	0.00	0.0003	0.0000	0.0012
Dactyloidae	*Anolis*	*cupreus*	0.00	0.00	0.0002	0.0000	0.1520
Dactyloidae	*Anolis*	*cybotes*	0.02	0.00	0.0011	0.0000	0.0676
Dactyloidae	*Anolis*	*distichus*	0.00	0.00	0.0000	0.0000	0.0042
Dactyloidae	*Anolis*	*etheridgei*	0.00	0.00	0.0000	0.0000	0.0002
Dactyloidae	*Anolis*	*hendersoni*	0.00	0.00	0.0000	0.0000	0.0004
Dactyloidae	*Anolis*	*humilis*	0.03	0.00	0.0026	0.0000	0.0815
Dactyloidae	*Anolis*	*insolitus*	1.00	0.00	0.0002	0.0000	0.0002
Dactyloidae	*Anolis*	*landestoyi*	0.00	0.00	0.0000	0.0000	0.0000
Dactyloidae	*Anolis*	*lemurinus*	0.02	0.00	0.0132	0.0000	0.5731
Dactyloidae	*Anolis*	*limifrons*	0.00	0.00	0.0000	0.0000	0.2125
Dactyloidae	*Anolis*	*lionotus*	0.02	0.00	0.0014	0.0000	0.0594
Dactyloidae	*Anolis*	*longitibialis*	0.00	0.00	0.0000	0.0000	0.0009
Dactyloidae	*Anolis*	*marcanoi*	0.00	0.00	0.0000	0.0000	0.0004
Dactyloidae	*Anolis*	*olssoni*	0.00	0.00	0.0000	0.0000	0.0309
Dactyloidae	*Anolis*	*placidus*	NA	NA	0.0000	0.0000	0.0000
Dactyloidae	*Anolis*	*ricordii*	0.00	0.00	0.0000	0.0000	0.0114
Dactyloidae	*Anolis*	*semilineatus*	0.00	0.00	0.0000	0.0000	0.0473
Dactyloidae	*Anolis*	*sheplani*	0.00	0.00	0.0000	0.0000	0.0001

Family	Genus	Species	Proportion < CT_{min}	Proportion < 0 °C	Area < CT_{min} (10^6 km^2)	Area < 0 °C (10^6 km^2)	Total area (10^6 km^2)
Dactyloidae	*Anolis*	*shrevei*	1.00	0.00	0.0004	0.0000	0.0004
Dactyloidae	*Anolis*	*singularis*	0.00	0.00	0.0000	0.0000	0.0000
Dactyloidae	*Anolis*	*strahmi*	0.00	0.00	0.0000	0.0000	0.0001
Dactyloidae	*Anolis*	*tropidolepis*	0.20	0.00	0.0006	0.0000	0.0031
Dactyloidae	*Anolis*	*whitemani*	0.00	0.00	0.0000	0.0000	0.0103
Scincidae	*Bassiana*	*duperreyi*	1.00	0.88	0.0289	0.0253	0.0289
Chamaeleonidae	*Chamaeleo*	*bitaeniatus*	0.01	0.00	0.0082	0.0000	0.7404
Chamaeleonidae	*Chamaeleo*	*dilepis*	0.24	0.01	2.6802	0.1400	11.0429
Chamaeleonidae	*Chamaeleo*	*ellioti*	0.00	0.00	0.0000	0.0000	0.9448
Chamaeleonidae	*Chamaeleo*	*hoehnelii*	0.05	0.00	0.0028	0.0000	0.0581
Chamaeleonidae	*Chamaeleo*	*jacksonii*	0.06	0.00	0.0018	0.0000	0.0285
Chamaeleonidae	*Chamaeleo*	*schubotzi*	0.56	0.24	0.0003	0.0001	0.0005
Teiidae	*Cnemidophorus*	*tigris*	0.99	0.51	1.9556	1.0018	1.9691
Eublepharidae	*Coleonyx*	*brevis*	1.00	0.32	0.5006	0.1600	0.5018
Eublepharidae	*Coleonyx*	*variegatus*	0.83	0.21	0.5278	0.1311	0.6337
Cordylidae	*Cordylus*	*cordylus*	0.96	0.18	0.2135	0.0403	0.2226
Cordylidae	*Cordylus*	*niger*	0.36	0.00	0.0018	0.0000	0.0049
Cordylidae	*Cordylus*	*oelofseni*	1.00	0.00	0.0003	0.0000	0.0003
Scincidae	*Ctenotus*	*regius*	1.00	0.00	1.5705	0.0000	1.5705
Scincidae	*Ctenotus*	*taeniolatus*	0.97	0.11	0.1493	0.0171	0.1543
Scincidae	*Ctenotus*	*uber*	1.00	0.00	0.0021	0.0000	0.0021
Iguanidae	*Dipsosaurus*	*dorsalis*	1.00	0.11	0.4551	0.0513	0.4551
Scincidae	*Egernia*	*cunninghami*	0.97	0.15	0.0958	0.0149	0.0983
Scincidae	*Egernia*	*saxatilis*	1.00	0.94	0.0184	0.0174	0.0184
Scincidae	*Egernia*	*striolata*	0.70	0.00	1.2007	0.0000	1.7217
Scincidae	*Egernia*	*whitii*	0.61	0.05	0.5240	0.0463	0.8552
Anguidae	*Elgaria*	*multicarinata*	0.56	0.28	0.1294	0.0661	0.2329
Lacertidae	*Eremias*	*multiocellata*	1.00	1.00	3.5235	3.5235	3.5235
Scincidae	*Eremiascincus*	*fasciolatus*	0.73	0.00	0.2857	0.0000	0.3926
Scincidae	*Eulamprus*	*kosciuskoi*	0.73	0.35	0.0261	0.0123	0.0356
Scincidae	*Eulamprus*	*quoyii*	0.69	0.02	0.6909	0.0187	0.9989
Scincidae	*Eulamprus*	*tympanum*	0.38	0.14	0.0943	0.0351	0.2478
Gekkonidae	*Gehyra*	*variegata*	0.58	0.00	0.6803	0.0000	1.1707
Gekkonidae	*Hemidactylus*	*frenatus*	0.22	0.00	1.6673	0.0000	7.4773
Gekkonidae	*Hemidactylus*	*turcicus*	0.76	0.06	1.7051	0.1321	2.2305
Scincidae	*Hemiergis*	*decresiensis*	1.00	0.00	0.0406	0.0000	0.0407
Gekkonidae	*Heteronotia*	*binoei*	0.84	0.00	1.9187	0.0000	2.2905

Family	Genus	Species	Proportion < CT_{min}	Proportion < 0 °C	Area < CT_{min} (10^6 km^2)	Area < 0 °C (10^6 km^2)	Total area (10^6 km^2)
Lacertidae	*Lacerta*	*agilis*	1.00	0.98	10.1618	9.9945	10.1618
Scincidae	*Lampropholis*	*delicata*	0.80	0.15	0.0765	0.0142	0.0961
Scincidae	*Lampropholis*	*guichenoti*	0.18	0.06	0.1283	0.0395	0.7060
Agamidae	*Laudakia*	*stellio*	1.00	0.35	0.4905	0.1696	0.4917
Gekkonidae	*Lepidodactylus*	*lugubris*	0.24	0.00	1.8065	0.0036	7.5835
Xantusiidae	*Lepidophyma*	*flavimaculatum*	0.02	0.00	0.0060	0.0000	0.3279
Liolaemidae	*Liolaemus*	*abaucan*	1.00	0.00	0.0011	0.0000	0.0011
Liolaemidae	*Liolaemus*	*albiceps*	1.00	1.00	0.0011	0.0011	0.0011
Liolaemidae	*Liolaemus*	*alticolor*	1.00	1.00	0.0479	0.0479	0.0479
Liolaemidae	*Liolaemus*	*altissimus*	1.00	0.85	0.0078	0.0066	0.0078
Liolaemidae	*Liolaemus*	*bellii*	1.00	0.85	0.0078	0.0066	0.0078
Liolaemidae	*Liolaemus*	*bibronii*	1.00	0.77	0.6101	0.4728	0.6101
Liolaemidae	*Liolaemus*	*bisignatus*	1.00	0.00	0.0813	0.0000	0.0817
Liolaemidae	*Liolaemus*	*boulengeri*	1.00	0.75	0.5668	0.4262	0.5668
Liolaemidae	*Liolaemus*	*canqueli*	1.00	1.00	0.0080	0.0080	0.0080
Liolaemidae	*Liolaemus*	*capillitas*	1.00	0.99	0.0066	0.0066	0.0066
Liolaemidae	*Liolaemus*	*chacoensis*	1.00	0.00	0.0355	0.0000	0.0355
Liolaemidae	*Liolaemus*	*chiliensis*	0.69	0.33	0.1401	0.0673	0.2029
Liolaemidae	*Liolaemus*	*constanzae*	1.00	0.82	0.0190	0.0155	0.0190
Liolaemidae	*Liolaemus*	*curis*	1.00	1.00	0.0002	0.0002	0.0002
Liolaemidae	*Liolaemus*	*cuyanus*	1.00	0.29	0.0946	0.0278	0.0946
Liolaemidae	*Liolaemus*	*cyanogaster*	1.00	0.20	0.1208	0.0246	0.1208
Liolaemidae	*Liolaemus*	*darwinii*	1.00	0.40	0.7572	0.3048	0.7572
Liolaemidae	*Liolaemus*	*dorbignyi*	1.00	0.98	0.0427	0.0419	0.0427
Liolaemidae	*Liolaemus*	*eleodori*	1.00	1.00	0.0009	0.0009	0.0009
Liolaemidae	*Liolaemus*	*elongatus*	1.00	1.00	0.2283	0.2280	0.2283
Liolaemidae	*Liolaemus*	*fabiani*	1.00	0.56	0.0025	0.0014	0.0025
Liolaemidae	*Liolaemus*	*fitzgeraldi*	1.00	0.98	0.0353	0.0347	0.0353
Liolaemidae	*Liolaemus*	*fitzingerii*	1.00	0.61	0.2226	0.1358	0.2226
Liolaemidae	*Liolaemus*	*fuscus*	0.84	0.03	0.0027	0.0001	0.0032
Liolaemidae	*Liolaemus*	*hellmichi*	1.00	0.00	0.0145	0.0000	0.0145
Liolaemidae	*Liolaemus*	*hernani*	1.00	0.90	0.0016	0.0015	0.0016
Liolaemidae	*Liolaemus*	*irregularis*	1.00	1.00	0.0091	0.0091	0.0091
Liolaemidae	*Liolaemus*	*jamesi*	1.00	0.99	0.0250	0.0248	0.0250
Liolaemidae	*Liolaemus*	*kingii*	1.00	0.79	0.2735	0.2156	0.2735
Liolaemidae	*Liolaemus*	*koslowskyi*	1.00	0.23	0.0287	0.0066	0.0287
Liolaemidae	*Liolaemus*	*kriegi*	1.00	1.00	0.0730	0.0730	0.0730

Family	Genus	Species	Proportion < CT_{min}	Proportion < 0 °C	Area < CT_{min} (10^6 km^2)	Area < 0 °C (10^6 km^2)	Total area (10^6 km^2)
Liolaemidae	*Liolaemus*	*laurenti*	1.00	0.02	0.0061	0.0001	0.0061
Liolaemidae	*Liolaemus*	*lemniscatus*	0.91	0.29	0.1485	0.0470	0.1627
Liolaemidae	*Liolaemus*	*leopardinus*	1.00	1.00	0.0004	0.0004	0.0004
Liolaemidae	*Liolaemus*	*lorenzmuelleri*	1.00	1.00	0.0001	0.0001	0.0001
Liolaemidae	*Liolaemus*	*melanops*	1.00	0.62	0.1332	0.0827	0.1332
Liolaemidae	*Liolaemus*	*monticola*	0.82	0.25	0.0198	0.0060	0.0241
Liolaemidae	*Liolaemus*	*multicolor*	1.00	1.00	0.0331	0.0331	0.0331
Liolaemidae	*Liolaemus*	*multimaculatus*	1.00	0.00	0.0099	0.0000	0.0099
Liolaemidae	*Liolaemus*	*nigromaculatus*	0.90	0.00	0.0731	0.0000	0.0817
Liolaemidae	*Liolaemus*	*nigroroseus*	1.00	0.45	0.0156	0.0070	0.0156
Liolaemidae	*Liolaemus*	*nigroviridis*	0.76	0.45	0.0119	0.0070	0.0156
Liolaemidae	*Liolaemus*	*nitidus*	0.45	0.18	0.0413	0.0171	0.0927
Liolaemidae	*Liolaemus*	*olongasta*	1.00	0.47	0.0275	0.0131	0.0275
Liolaemidae	*Liolaemus*	*ornatus*	1.00	0.98	0.1279	0.1254	0.1279
Liolaemidae	*Liolaemus*	*petrophilus*	1.00	0.97	0.1681	0.1638	0.1681
Liolaemidae	*Liolaemus*	*pictus*	1.00	0.38	0.1747	0.0655	0.1747
Liolaemidae	*Liolaemus*	*platei*	1.00	0.18	0.0746	0.0131	0.0746
Liolaemidae	*Liolaemus*	*poecilochromus*	1.00	1.00	0.0048	0.0048	0.0048
Liolaemidae	*Liolaemus*	*pseudoanomalus*	1.00	0.31	0.0226	0.0070	0.0226
Liolaemidae	*Liolaemus*	*pseudolemniscatus*	1.00	0.03	0.0224	0.0007	0.0224
Liolaemidae	*Liolaemus*	*quilmes*	1.00	0.40	0.0094	0.0037	0.0094
Liolaemidae	*Liolaemus*	*riojanus*	1.00	0.02	0.0174	0.0003	0.0174
Liolaemidae	*Liolaemus*	*robertmertensi*	1.00	0.16	0.0215	0.0035	0.0215
Liolaemidae	*Liolaemus*	*rothi*	1.00	1.00	0.1269	0.1269	0.1269
Liolaemidae	*Liolaemus*	*salinicola*	1.00	0.05	0.0057	0.0003	0.0057
Liolaemidae	*Liolaemus*	*scapularis*	1.00	0.00	0.0052	0.0000	0.0052
Liolaemidae	*Liolaemus*	*schroederi*	0.92	0.29	0.0536	0.0171	0.0580
Liolaemidae	*Liolaemus*	*telsen*	1.00	0.99	0.0227	0.0225	0.0227
Liolaemidae	*Liolaemus*	*tenuis*	0.75	0.26	0.1487	0.0522	0.1977
Liolaemidae	*Liolaemus*	*vallecurensis*	1.00	1.00	0.0285	0.0285	0.0285
Liolaemidae	*Liolaemus*	*walkeri*	1.00	0.67	0.0365	0.0245	0.0365
Liolaemidae	*Liolaemus*	*xanthoviridis*	1.00	0.00	0.0035	0.0000	0.0035
Agamidae	*Moloch*	*horridus*	0.00	0.00	0.0045	0.0000	2.9143
Scincidae	*Nannoscincus*	*maccoyi*	1.00	0.98	0.0155	0.0152	0.0155
Scincidae	*Niveoscincus*	*metallicus*	0.41	0.07	0.0429	0.0071	0.1056
Scincidae	*Niveoscincus*	*ocellatus*	0.67	0.12	0.0342	0.0060	0.0514
Phrynosomatidae	*Phrynosoma*	*cornutum*	0.99	0.58	1.7404	1.0145	1.7573

Family	Genus	Species	Proportion < CT_{min}	Proportion < 0 °C	Area < CT_{min} (10^6 km^2)	Area < 0 °C (10^6 km^2)	Total area (10^6 km^2)
Phrynosomatidae	*Phrynosoma*	*douglassii*	1.00	1.00	0.2099	0.2099	0.2099
Liolaemidae	*Phymaturus*	*antofagastensis*	1.00	1.00	0.0025	0.0025	0.0025
Liolaemidae	*Phymaturus*	*dorsimaculatus*	NA	NA	0.0000	0.0000	0.0000
Liolaemidae	*Phymaturus*	*indistinctus*	1.00	1.00	0.0029	0.0029	0.0029
Liolaemidae	*Phymaturus*	*palluma*	1.00	0.95	0.1485	0.1417	0.1485
Liolaemidae	*Phymaturus*	*patagonicus*	1.00	0.66	0.0675	0.0445	0.0675
Liolaemidae	*Phymaturus*	*somuncurensis*	1.00	1.00	0.0113	0.0113	0.0113
Liolaemidae	*Phymaturus*	*tenebrosus*	1.00	1.00	0.0032	0.0032	0.0032
Liolaemidae	*Phymaturus*	*zapalensis*	1.00	1.00	0.0089	0.0089	0.0089
Cordylidae	*Platysaurus*	*intermedius*	0.97	0.00	0.3252	0.0000	0.3363
Scincidae	*Plestiodon*	*elegans*	0.92	0.26	1.6206	0.4597	1.7696
Scincidae	*Plestiodon*	*gilberti*	1.00	0.32	0.2161	0.0699	0.2161
Scincidae	*Pseudemoia*	*entrecasteauxii*	0.38	0.10	0.1699	0.0425	0.4462
Scincidae	*Pseudemoia*	*spenceri*	1.00	0.91	0.0215	0.0195	0.0215
Cordylidae	*Pseudocordylus*	*melanotus*	1.00	0.76	0.0235	0.0180	0.0235
Agamidae	*Rankinia*	*diemensis*	0.64	0.13	0.2247	0.0448	0.3527
Scincidae	*Saproscincus*	*mustelina*	1.00	0.54	0.0207	0.0112	0.0207
Phrynosomatidae	*Sceloporus*	*graciosus*	1.00	0.95	1.3667	1.2987	1.3697
Phrynosomatidae	*Sceloporus*	*malachiticus*	0.05	0.00	0.0027	0.0000	0.0541
Phrynosomatidae	*Sceloporus*	*occidentalis*	0.96	0.74	0.9918	0.7613	1.0304
Phrynosomatidae	*Sceloporus*	*undulatus*	1.00	0.72	3.9060	2.8258	3.9060
Phrynosomatidae	*Sceloporus*	*variabilis*	0.48	0.00	0.4679	0.0015	0.9838
Scincidae	*Sphenomorphus*	*incognitus*	0.95	0.08	0.3273	0.0276	0.3442
Scincidae	*Sphenomorphus*	*taiwanensis*	1.00	0.30	0.0031	0.0009	0.0031
Lacertidae	*Takydromus*	*formosanus*	0.00	0.00	0.0000	0.0000	0.0153
Lacertidae	*Takydromus*	*hsuehshanensis*	0.71	0.26	0.0015	0.0005	0.0021
Lacertidae	*Takydromus*	*septentrionalis*	0.88	0.47	2.0398	1.0952	2.3084
Lacertidae	*Takydromus*	*sexlineatus*	0.15	0.01	0.6763	0.0312	4.4720
Lacertidae	*Takydromus*	*stejnegeri*	0.00	0.00	0.0000	0.0000	0.0188
Phyllodactylidae	*Tarentola*	*boettgeri*	0.15	0.00	0.0004	0.0000	0.0024
Sphaerodactylidae	*Teratoscincus*	*przewalskii*	1.00	1.00	0.8615	0.8615	0.8615
Scincidae	*Tiliqua*	*nigrolutea*	1.00	0.95	0.0207	0.0196	0.0207
Scincidae	*Tiliqua*	*rugosa*	0.08	0.00	0.2492	0.0000	2.9500
Agamidae	*Trapelus*	*savignii*	1.00	0.00	0.0160	0.0000	0.0160
Varanidae	*Varanus*	*varius*	0.65	0.01	0.6258	0.0053	0.9649
Xantusiidae	*Xantusia*	*vigilis*	0.85	0.47	0.1969	0.1083	0.2312
Lacertidae	*Zootoca*	*vivipara*	0.99	0.96	17.5053	17.0789	17.7336

Supplementary Table 4 | Outlier analysis comparing the cold-climate mismatch of the American green anole (*Anolis carolinensis*) with that of other lizards. For each metric, we calculated the Z-score of *A. carolinensis* relative to three comparison groups: *Anolis* species (n = 35), exclusively arboreal lizards (n = 25) and all lizards in our dataset (n = 180). The Z-score is the number of standard deviations by which *A. carolinensis* lies above (positive values) or below (negative values) the group mean. "Below CT_{min}" and "below 0 °C" refer to areas where the minimum temperature of the coldest month (BIO6) falls below each species' critical thermal minimum (6.8 °C for *A. carolinensis*, the lowest population-average CT_{min} across its range) or below freezing. Because areas span several orders of magnitude, their Z-scores were calculated on $\log_{10}(\text{area} + 1\ km^2)$. Because many species have no area below a threshold, we also report the rank of *A. carolinensis* within each group (1 = largest value; ties share the highest rank), as rank / number of species with data. Proportions could not be calculated for *A. placidus* and *Phymaturus dorsimaculatus*. *Anolis carolinensis* is included in all three groups, and it is the only *Anolis* whose range extends below 0 °C, so its Z-scores for these metrics relative to *Anolis* reflect a single non-zero value.

	***Anolis* genus**		**Arboreal lizards**		**All lizards**	
Metric	**Z-score**	**Rank**	**Z-score**	**Rank**	**Z-score**	**Rank**
Proportion of range below CT_{min}	1.97	5 / 34	2.13	3 / 24	0.43	103 / 178
Proportion of range below 0 °C	5.66	1 / 34	3.46	1 / 24	-0.19	76 / 178
Area of range below CT_{min} ($\log_{10}$ km^2)	2.93	1 / 35	2.82	1 / 25	1.13	17 / 180
Area of range below 0 °C ($\log_{10}$ km^2)	5.75	1 / 35	4.37	1 / 25	1.23	14 / 180

Supplementary Table 5 | Temperature, not photoperiod, triggers the retreat of green anoles into the leaf litter. Results of a generalized linear model with a Bernoulli error structure (logit link) examining how the probability of observing American green anoles (*Anolis carolinensis*) in the vegetation during night surveys, rather than retreated into the leaf litter, changed with the average canopy operative temperature during the survey period (1–2 h after sunset; T), photoperiod (D; hours between sunrise and sunset in Atlanta on the day of the survey) and their interaction. Temperature and photoperiod were centered on their means across surveys, so the intercept (α) and the temperature effect (β_T) apply at an average photoperiod, and the photoperiod effect (β_D) at an average temperature. The *z* statistic tests whether each coefficient differs from zero (Wald test). Likelihood-ratio tests (LRT) compare the full model with a model without the interaction (1 df) or without a predictor and its interaction (2 df). Tjur's R^2 is the difference between the mean predicted observation probability for surveys with and without lizards in the vegetation. Temperature thresholds are the canopy temperatures at which the predicted observation probability equals each value at the average photoperiod, with 95% confidence intervals obtained with the delta method. The behavioral retreating threshold temperature (T_{ret}), below which green anoles retreat into the leaf litter, is defined as the temperature at which observation probability falls to 0.20.

Parameter estimates

Parameter	Symbol	Estimate	Std. Error	z	p-value	LRT χ^2	LRT df	LRT p-value
Intercept	α	0.323	0.557	0.58	0.5620	–	–	–
Operative temperature (°C, centered)	β_T	0.346	0.127	2.74	0.0062	16.80	2	0.0002
Photoperiod (h, centered)	β_D	0.735	0.877	0.84	0.4016	0.98	2	0.6139
Temperature × Photoperiod	$\beta_{T\cdot D}$	0.049	0.182	0.27	0.7861	0.08	1	0.7769

Model performance metrics

Surveys	34 (19 with lizards; 15 without)
Mean canopy temperature (°C)	13.10
Mean photoperiod (h)	10.87
Log-likelihood	-12.53
AIC	33.1
Tjur's R^2	0.51

Inferred operative temperature thresholds (at the average photoperiod)

Observation probability	Operative temperature in canopy (°C)	95% CI (°C)
0.01	-1.10	-11.99 to 9.79
0.05	3.67	-4.02 to 11.36
0.10	5.82	-0.48 to 12.13
0.20	8.17	3.24 to 13.09
0.25	9.00	4.51 to 13.48
0.50	12.17	8.89 to 15.44

Supplementary Table 6 | Average (± SD across operative temperature models, OTMs) and absolute minimum nightly operative temperatures recorded by OTMs in the canopy and in the leaf litter at each location during January 2026, the coldest month of the year. “Buffer” is the difference between the average minimum temperatures in the leaf litter and in the canopy. We recorded temperatures with 22 canopy and 17 leaf-litter OTMs across the five locations, after excluding two leaf-litter OTMs in Atlanta that became exposed during deployment.

		Minimum operative temperatures				
		Canopy (°C)		Leaf litter (°C)		
Location	**Lat. (°N)**	**Mean ± SD**	**Min.**	**Mean ± SD**	**Min.**	**Buffer (°C)**
Miami, FL (MIA)	25.76	14.9 ± 0.3	5.0	19.2 ± 0.5	13.5	4.3
Port Saint Lucie, FL (PSL)	27.30	12.5 ± 0.2	3.5	16.2 ± 0.5	10.5	3.6
Orlando, FL (ORL)	28.54	8.5 ± 0.3	-1.5	14.6 ± 1.6	7.5	6.1
Gainesville, FL (GNV)	29.65	6.8 ± 0.1	-2.5	11.6 ± 1.0	5.0	4.8
Atlanta, GA (ATL)	33.75	1.7 ± 0.4	-11.0	6.6 ± 1.3	-2.5	4.9

Supplementary Table 7 | Results of a linear mixed model examining how nightly minimum operative temperatures recorded by operative temperature models (OTMs) in January changed with latitude and microhabitat (canopy vs. leaf litter). M is an indicator variable for microhabitat (M = 0 for canopy, M = 1 for leaf litter). Latitude (L) was centered on the mean latitude of the five locations (29.0°N), so α is the canopy minimum temperature at that latitude, and β_M is the average buffering provided by the leaf litter (leaf-litter minus canopy minimum temperature) at that latitude. The model includes random intercepts for each OTM, location and night within location. Each fixed effect is tested against zero with a t-test whose standard error and denominator degrees of freedom (df) use the Kenward–Roger approximation; df vary among terms because latitude varies only among the five locations, whereas microhabitat varies among OTMs within locations. The latitudinal slopes give the rate at which minimum temperatures decline per degree of latitude in each microhabitat (β_L for the canopy and $\beta_L + \beta_{L \cdot M}$ for the leaf litter). Two leaf-litter OTMs in Atlanta that became exposed during deployment were excluded.

Parameter estimates

Parameter	Symbol	Estimate	Std. Error	df	t	p-value
Intercept	α	8.829	0.431	3.7	20.48	< 0.0001
Latitude (centered)	β_L	-1.632	0.151	3.0	-10.79	0.0018
Leaf litter	β_M	4.732	0.301	32.4	15.70	< 0.0001
Latitude × Leaf litter	$\beta_{L \cdot M}$	0.053	0.085	32.3	0.63	0.5362

Latitudinal slopes by microhabitat

Microhabitat	Slope (°C per ° latitude)	Std. Error	95% CI	df	t	p-value
Canopy (β_L)	-1.632	0.151	-2.117 to -1.147	3.0	-10.79	0.0018
Leaf litter ($\beta_L + \beta_{L \cdot M}$)	-1.578	0.152	-2.058 to -1.099	3.0	-10.41	0.0018

Random effects and variance components

Component	Symbol	SD
OTM	σ_O	0.751
Location	σ_S	0.193
Night (within location)	σ_N	4.467
Residual	σ	1.593

Model information

Observations	1180 nightly minima from 39 OTMs (22 canopy; 17 leaf litter) at 5 locations
Log-likelihood (REML)	-2556.75
AIC (REML)	5129.5

Supplementary Table 8 | Validation of the biophysical models. Agreement between the minimum operative temperatures predicted by our biophysical models and the mean daily minimum temperatures recorded by operative temperature models (OTMs) between 11 and 19 January at five locations, in the canopy (1.8 m) and in the leaf litter (8 cm). Bias is the mean difference between predicted and observed temperatures (positive values indicate that the model runs warm), MAE the mean absolute error and RMSE the root mean square error across individual OTMs. RMSE is also expressed as a percentage of the range of observed temperatures, and calculated across site means (i.e., averaging all OTMs per site and microhabitat first), which weights each site equally.

Microhabitat	Sites	OTMs	Bias (°C)	MAE (°C)	RMSE (°C)	RMSE (% of observed range)	RMSE, site means (°C)
Canopy	5	22	0.45	0.76	0.88	5.5	0.97
Leaf Litter	5	17	-0.41	1.44	1.76	12.1	1.12

Supplementary Table 9 | Results of the mechanistic envelope models. Combination of microclimate (depth in the leaf litter) and forest cover threshold that yielded the highest true skill statistic (TSS) for each spatial extent. Spatial extents are defined by the minimum temperature at 1.8 m in the canopy (T_{min}): areas where it falls below 0 °C, below CT_{min} (6.8 °C), below T_{ret} (8.17 °C), and the entire study area.

Spatial extent	Leaf-litter depth (cm)	Forest cover threshold (%)	TSS
Canopy T_{min} < 0 °C	4	35	0.812
Canopy T_{min} < CT_{min}	3	20	0.777
Canopy T_{min} < T_{ret}	3	20	0.766
Entire area	3	20	0.750

Supplementary Table 10 | Predictive capacity of each variable on its own in the mechanistic envelope models. Values are the highest true skill statistic (TSS) achieved by forest cover alone (cells suitable if forest cover exceeds a threshold), by temperature alone (cells suitable if the minimum annual body temperature in the canopy, or in the best-performing microclimate, is above 0 °C) and by the combination of temperature and forest cover, with the threshold or microclimate that achieved it in parentheses. In areas where canopy temperatures fall below 0 °C, canopy temperatures predict all cells to be unsuitable by definition (TSS = 0).

	Forest cover alone	Temperature alone		Temperature and forest cover
Spatial extent	Forest cover	Canopy (1.8 m)	Best microclimate	Best microclimate + forest cover
Canopy T_{min} < 0 °C	0.445 (> 45%)	0.000	0.670 (4 cm deep in leaf litter)	0.812 (4 cm deep in leaf litter; > 35%)
Canopy T_{min} < CT_{min}	0.403 (> 45%)	0.463	0.634 (2 cm deep in leaf litter)	0.777 (3 cm deep in leaf litter; > 20%)
Canopy T_{min} < T_{ret}	0.403 (> 45%)	0.449	0.613 (2 cm deep in leaf litter)	0.766 (3 cm deep in leaf litter; > 20%)
Entire area	0.387 (> 40%)	0.446	0.591 (2 cm deep in leaf litter)	0.750 (3 cm deep in leaf litter; > 20%)

Values are the highest TSS achieved by each type of model, with the threshold or microclimate that achieved it in parentheses. In areas where canopy temperatures fall below 0 °C, canopy temperatures predict all cells to be unsuitable by definition (TSS = 0).

Supplementary Table 11 | Parameters of the biophysical models (Supplementary Note 1).

Parameter	Symbol	Value	Units	Source
Day of the year	DOY	15 (15 January)	–	–
Eccentricity of the Earth's orbit	e	0.0167	–	–
Extraterrestrial solar constant	E_{sc}	1.36	kW m^{-2}	ref. 3
Atmospheric transmittance coefficient	τ	0.83	–	ref. 3
Canopy cooling coefficient	b_c	0.007	°C $(W\ m^{-2})^{-1}$	refs. 4–6
Surface heating coefficient	b_s	0.02	°C $(W\ m^{-2})^{-1}$	refs. 3,7
Reference height of air temperature	z_{ref}	1.8	m	WorldClim
Roughness length	z_0	0.01	m	–
Thermal conductivity of leaf litter	k	0.05	W m^{-1} K^{-1}	refs. 3,8
Bulk density of leaf litter	ρ	150	kg m^{-3}	refs. 9,10
Specific heat capacity of leaf litter	c_p	2500	J kg^{-1} K^{-1}	refs. 11,12
Body mass of a green anole	m	3.1	g	–
Wind speed	u	0.1	m s^{-1}	–
Conductive heat transfer coefficient	h_k	2.5	W m^{-2} K^{-1}	–
Stefan-Boltzmann constant	σ	5.670374×10^{-8}	W m^{-2} K^{-4}	–
Emissivity of lizard skin	ϵ_{lizard}	0.95	–	–
Emissivity of the night sky	ϵ_{sky}	0.75	–	–
Sky and ground view factors	F_{sky}, F_{ground}	0.5	–	–

Supplementary References

1. Fick, S. E. & Hijmans, R. J. WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas. *Int. J. Climatol.* **37**, 4302–4315 (2017).
2. Buckley, L. B. et al. TrenchR: an R package for modular and accessible microclimate and biophysical ecology. *PLoS Clim.* **2**, e0000139 (2023).
3. Campbell, G. S. & Norman, J. M. *An Introduction to Environmental Biophysics* 2nd edn (Springer, 1998).
4. De Frenne, P. et al. Microclimate moderates plant responses to macroclimate warming. *Proc. Natl Acad. Sci. USA* **110**, 18561–18565 (2013).
5. De Frenne, P. et al. Global buffering of temperatures under forest canopies. *Nat. Ecol. Evol.* **3**, 744–749 (2019).
6. Zellweger, F. et al. Forest microclimate dynamics drive plant responses to warming. *Science* **368**, 772–775 (2020).
7. Maclean, I. M. D., Mosedale, J. R. & Bennie, J. J. Microclima: an R package for modelling meso- and microclimate. *Methods Ecol. Evol.* **10**, 280–290 (2019).
8. Oke, T. R. *Boundary Layer Climates* 2nd edn (Routledge, 1987).
9. Gaudinski, J. B. et al. Soil carbon cycling in a temperate forest: radiocarbon-based estimates of residence times, sequestration rates and partitioning of fluxes. *Biogeochemistry* **51**, 33–69 (2000).
10. Sato et al. (2004) [full reference to be added].
11. De Vries, D. A. Thermal properties of soils. In *Physics of Plant Environment* (ed. van Wijk, W. R.) (North-Holland, 1963).
12. Bonan, G. *Climate Change and Terrestrial Ecosystem Modeling* (Cambridge Univ. Press, 2019).